SIMATS ENGINEERING
ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I **Shaik Mohammed Waseem Rayan (192373004)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Shaik Mohammed Waseem Rayan

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground Aim

To prove that the ground is wet using the **Resolution Algorithm**. **Knowledge Base**

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet.

Step 1: Propositional Representation

Let

- R = It is raining
- W = Ground is wet

Knowledge Base:

1. $R \rightarrow W$
2. R

Goal: W

Step 2: Convert into CNF

Using $P \rightarrow Q \equiv \neg P \lor Q$

- C1 : $\neg R \lor W$
- C2 : R

Step 3: Negate the Goal

Goal = W Negated Goal:

- C3 : $\neg W$

Step 4: Resolution

Resolve C1 and C2:

(¬R ∨ W), (R)

⇒ C4 : W

Resolve C4 and C3:

1. , (¬W)

⇒ □ (Empty Clause)

Conclusion

Since the **Empty Clause (□)** is obtained, the negation of the goal is false.

Therefore, the ground is wet is proved.

Question 2 – Student Assignment Submission

Aim

To prove that Rahul receives internal marks using Resolution.

Knowledge Base

- If a student submits the assignment, then the student receives internal marks.
- Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks.

Step 1: Propositional Representation

- S = Rahul submitted assignment
- M = Rahul receives internal marks

Knowledge Base:

1. S → M
2. S

Goal: M

Step 2: CNF

- C1 : ¬S ∨ M
- C2 : S

Step 3: Negate Goal

- C3 : ¬M

Step 4: Resolution

Resolve C1 and C2:

(¬S ∨ M), (S)

⇒ C4 : M

Resolve C4 and C3:

1. , (¬M)

⇒ □

Conclusion

Empty Clause is obtained.

Hence, Rahul receives internal marks is proved.

Question 3 – Library Membership

Aim

To prove that Priya can borrow books using Resolution.

Knowledge Base

- If a person is a library member, then the person can borrow books.
- Priya is a library member.

Goal

Prove that Priya can borrow books.

Step 1: Propositional Representation

- L = Priya is a library member
- B = Priya can borrow books

Knowledge Base:

1. L → B
2. L

Goal: B

Step 2: CNF

- $C1 : \neg L \lor B$
- $C2 : L$

Step 3: Negate Goal

- $C3 : \neg B$

Step 4: Resolution

Resolve C1 and C2:

$(\neg L \lor B), (L)$

$\Rightarrow C4 : B$

Resolve C4 and C3:

1. , $(\neg B)$

$\Rightarrow \square$

No further clauses can be generated.

Conclusion

Empty Clause is derived.

Therefore, Priya can borrow books is proved.

Question 4 – Placement Eligibility

Aim

To prove that Arun is eligible for placement using Resolution.

Knowledge Base

- If a student clears the aptitude test, then the student is eligible for placement.
- Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement.

Step 1: Propositional Representation

- A = Arun cleared aptitude test
- P = Arun is eligible for

placement Knowledge Base:

1. $A \rightarrow P$

2. A

Goal: P

Step 2: CNF

- C1 : ¬A ∨ P
- C2 : A

Step 3: Negate Goal

- C3 : ¬P

Step 4: Resolution

Resolve C1 and C2:

(¬A ∨ P), (A)

⇒ C4 : P

Resolve C4 and C3:

1. , (¬P)

⇒ □

Resolution Tree

```
¬A ∨ P
  |
  | Resolve with A
  |
  P
  |
  | Resolve with ¬P
  |
  □
```

Conclusion

Empty Clause is obtained.

Hence, Arun is eligible for placement is proved.

Question 5 – Access Control System

Aim

To prove that the user is granted access using Resolution.

Knowledge Base

- If a user enters the correct password, then the user is authenticated.
- If a user is authenticated, then the user is granted access.
- The user entered the correct password.

Goal

Prove that the user is granted access.

Step 1: Propositional Representation

- P = Correct password entered
- A = User authenticated
- G = User granted access

Knowledge Base:

1. $P \rightarrow A$
2. $A \rightarrow G$
3. P

Goal: G

Step 2: CNF

- C1 : $\neg P \lor A$
- C2 : $\neg A \lor G$
- C3 : P

Step 3: Negate Goal

- C4 : $\neg G$

Step 4: Resolution

Resolution 1

$(\neg P \lor A), (P)$

$\Rightarrow$ C5 : A

Resolution 2

$(\neg A \lor G), (A)$

⇒ C6 : G

Resolution 3

(G), (¬G)

⇒ □

Conclusion

Empty Clause is obtained.

Therefore, the user is granted access is proved using the Resolution Algorithm.

Overall Result

All the given goals are successfully proved using the **Resolution Algorithm**, and in each case the derivation of the **Empty Clause (□)** confirms that the goal logically follows from the Knowledge Base. This demonstrates the use of **resolution-based automated reasoning in Artificial Intelligence**.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation